ATLAS – uGDX

School of Technology

B. Tech in CS AI & ML

Academic Year:2026-27

BTECH CAP 501

PROJECT LIFE CYCLE MANAGEMENT

PROJECT REPORT

PROJECT TITLE

Project Initiation, Feasibility, and Requirements

Student Name		**Roll No.**	
Branch / Section		Academic Year	
Project Title		Project Domain	
Faculty Mentor		Date of Submission	

(FIRST PAGE)

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Purpose of the Report

This report is to be prepared individually based on the student's selected project. It is designed to demonstrate practical application of all concepts covered in Unit 1. At this stage, students are not required to build a complete machine learning model. They must initiate and analyse their own proposed AI/ML project by defining the problem, studying stakeholders, identifying requirements, comparing project life-cycle frameworks, and conducting feasibility analysis.

Unit 1: Concepts Covered

1.1 The AI and ML Project Life Cycle and why AI/ML projects differ from conventional software projects
1.2 AI Project Life Cycle Frameworks: CRISP-DM, Microsoft TDSP, Google ML Life Cycle
1.3 Problem Scoping and Feasibility: technical, economic, operational, ethical
1.4 Requirements Engineering for AI: functional, non-functional, data, ethical requirements
1.5 Stakeholder Management in AI: data subjects, regulators, affected communities, adversarial stakeholders

Unit 2: Concepts Covered

2.1 Data as the critical resource: Provenance, lineage and versioning.
2.2 The five data quality dimensions and their ML failure modes.

General Instructions

1. Prepare the report using your own selected project topic.
2. Every answer must be connected to your project; avoid generic textbook/Internet answers.
3. State assumptions wherever exact information is not available.
4. Use tables, diagrams and flowcharts where appropriate.
5. A complete ML implementation is not compulsory for this Unit 1 & 2 report.
6. Use references for external facts, datasets, regulations and claims.
7. Suggested report length: Minimum 20 –25 pages excluding cover page and references.

Chapter 1. Project Overview and Problem Scoping

1.1 Project Title

Write a clear and specific title.

1.2 Background and Context

Explain the real-world context and current situation in which the problem exists.

1.3 Problem Statement

Clearly define the specific problem. Describe the problem before proposing an algorithm.

1.4 Proposed AI/ML Solution

Explain how AI or machine learning could help address the identified problem.

1.5 Success Criteria

Define technical and practical/business criteria that will indicate project success. Your answers should be based on the following parameters:

Question	Project-Specific Answer
What is the input?	
What is the expected output?	
Who will use the output?	
How will the output help solve the problem?	

Type of Success	Success Criterion	How It Will Be Measured

Chapter 2. AI/ML Project Life Cycle and Difference from Conventional Software

2.1 Proposed Life Cycle of Your Project

Draw and describe the stages of your project from problem definition to deployment and monitoring. Show feedback loops where later stages may require returning to earlier stages.

2.2 Why Your Project Is an AI/ML Project

Explain the role of data, uncertainty, experimentation, model evaluation, monitoring and possible retraining.

Aspect	Your AI/ML Project
Primary input	
Expected behaviour	
Role of data	
Testing/evaluation	
Changes after deployment	
Monitoring requirement	

2.3 ML Project vs ML Pipeline

Identify activities that could be automated and decisions that require human judgement and accountability in your project.

Possible Automated Pipeline Activities	Human Decisions and Accountability

Chapter 3. AI Project Life Cycle Frameworks

Justify which life cycle framework is best suitable for your project out of the listed frameworks:

3.1 CRISP-DM Mapping

CRISP-DM Phase	How This Phase Applies to My Project
Business Understanding	
Data Understanding	
Data Preparation	
Modelling	
Evaluation	
Deployment	

3.2 Microsoft TDSP Mapping

TDSP Stage	Application to My Project
Business Understanding	
Data Acquisition and Understanding	
Modelling	
Deployment	
Customer Acceptance	

3.3 Google ML Life Cycle / Pipeline Perspective

Identify the stages that could form the ML pipeline for your project and explain what important decisions remain outside the automated pipeline.

Pipeline Stage	Possible Automation	Human Decision Required

Final framework choice and justification:

Chapter 4. Problem Scoping and Feasibility Analysis

Analyse technical, economic, operational and ethical feasibility. Identify evidence, risks, limitations and conditions required for success.

4.1 Technical Feasibility

Analyse data sufficiency, model capability, and latency/hardware capability.

Technical Aspect	Analysis for My Project	Risk / Limitation	Possible Mitigation
Data sufficiency			
Model capability			
Latency / hardware capability			

4.2 Economic Feasibility

Cost / Benefit Item	Description	Estimated Level (Low/Medium/High)	Justification
Data acquisition			
Data cleaning/labelling			
Computing/infrastructure			
Development effort			
Maintenance/retraining			
Expected benefit/value			

Economic feasibility conclusion:

4.3 Operational Feasibility

Operational Question	Analysis for My Project
Who will operate the system?	
Who will maintain the data?	
Who will monitor the model?	
Who will handle retraining?	
What skills/infrastructure are required?	
Can the organization sustain it over time?	

Operational feasibility conclusion:

4.4 Ethical Feasibility

Identify potential harms and determine whether the project should proceed and under what conditions. Ethical feasibility is not simply a trade-off against cost or performance.

Ethical Issue	Who May Be Affected?	Potential Harm	Requirement & Enforcement in Project
Privacy and data use			
Bias and fairness			
Transparency / explainability			
Misuse or harmful decisions			
Human oversight and accountability			

Ethical feasibility conclusion:

Chapter 5. Requirements Engineering for AI

Requirements must be specific to your project. Avoid vague statements such as 'the system should be fast' unless measurable targets are included.

5.1 Functional Requirements

What should the system do?

Aspect	Functional Requirement	Priority	Acceptance / Test Criterion
FR1			
FR2			
FR3			
FR4			
FR5			

5.2 Non-Functional Requirements

Specify quality, performance, reliability, scalability, maintainability or usability constraints.

Aspect	Non-Functional Requirement	Target / Constraint	How It Will Be Verified
NFR1			
NFR2			
NFR3			
NFR4			
NFR5			

5.3 Data Requirements

Data Requirement	Project-Specific Specification
Required data sources	
Required features/variables	
Data volume/availability	
Data format and integration	

Data quality	
Labelling requirements	
Data update frequency and retention	

Chapter 6. Stakeholder Management in AI

6.1 Stakeholder Identification

Identify stakeholders broadly; do not limit analysis to the project team or direct users.

Stakeholder	Category	Interest / Role	Influence	Potential Concern

6.2 Stakeholder Categories to Consider

- Model/system users
- Business owners or decision-makers
- Data owners
- Data subjects
- Regulators/compliance authorities
- Affected communities
- Auditors/reviewers
- Adversarial stakeholders

6.3 Stakeholder Conflict Analysis

Identify at least three possible conflicts among stakeholders with respect to your project.

Stakeholder 1	Stakeholder 2	Conflict	How the Project Will Address It

6.4 Data Subjects and Affected Communities

Explain who provides data to the system and who may be affected by its predictions or decisions. These groups may not be the same.

6.5 Adversarial Stakeholders

Identify possible individuals or groups who could intentionally misuse, manipulate, attack or exploit the system or its data.

Possible Adversarial Actor	Possible Action / Threat	Potential Impact	Basic Mitigation

Chapter 7. Dataset Overview

7.1 Data Required for the Project

Dataset/Data Item	Purpose in Project	Format	Expected Size	Availability Status

7.2 Data Provenance

Dataset	Source	Data Owner	Collected By	Collection Period	Collection Method	Permission/Access Status

7.3 Data Lineage Diagram

Create and insert a project-specific flow diagram.

Suggested flow: Data Source → Raw Data → Cleaning/Validation → Transformation → Feature Creation → Final Dataset

Explain how this lineage record would help reproduce a past result or investigate an incorrect result.

7.4 Data Transformation Log

Explain this step if in case the data collection and processing steps are completed for your project.

Step No.	Input Data	Operation Performed	Reason	Output Data	Tool/Method Used

Chapter 8. Final Project Initiation Summary

Item	Final Summary
Problem being addressed	
Proposed AI/ML approach	
Primary stakeholders	
Major data requirement	
Most important feasibility risk	
Most important ethical concern	
Final project recommendation	

Chapter 9. Preliminary Next Steps

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10. References

List all books, articles, websites, reports, datasets, policies or other sources used. Use IEEE citation style.

Appendix (Optional)

Additional stakeholder maps, feasibility calculations, diagrams, requirement documents or supporting material may be included.

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